Homework 3

Due Wednesday, May 20

You will use Python to repeat a similar exercise we did in class. In class we used data on 'Bitcoin' and 'Ether' cryptocurrencies. In the homework, you will use data for 'Cardano' and 'Dogecoin' cryptocurrencies. The ticker names for these two cryptocurrencies are 'ADA-USD' and 'DOGE-USD'. You will need these two ticker names to download data from yFinance.

1. Download data from yFinance for Cardano and Dogecoin, from January 1, 2020 until May 10, 2026.

2. Compute daily returns for these two cryptocurrencies.

3. Compute means, standard deviations and correlation for these two cryptocurrencies.

4. Construct 11 portfolios with the following allocations in the two cryptocurrencies:

   Portfolio 1: [0.0 , 1.0]
   Portfolio 2: [0.1 , 0.9]
   Portfolio 3: [0.2 , 0.8]
   Portfolio 4: [0.3 , 0.7]
   Portfolio 5: [0.4 , 0.6]
   Portfolio 6: [0.5 , 0.5]
   Portfolio 7: [0.6 , 0.4]
   Portfolio 8: [0.7 , 0.3]
   Portfolio 9: [0.8 , 0.2]
   Portfolio 10: [0.9 , 0.1]
   Portfolio 11: [1.0 , 0.0]

   For example, the first portfolio contains only Dogecoin. The second portfolio contains 10% Cardano and 90% Dogecoin. For each of the

11 portfolios, calculate (a) the average return; (b) the standard deviation or returns; (c) the Sharpe ratio.

5. Plot in a graph the standard deviation of each portfolio against the average return.

6. Plot in another graph the Sharpe ratio of each portfolio against the share of the portfolio allocated to Cardano.

Please submit two the notebook file containing the codes by email: quadrini@usc.edu. Note that you need to download the .ipynb file from Colab.